

Section 7 Practice Problem: Solutions

Exercise Programme Experiment

(i) Assignment mechanism and number of valid assignments.

The assignment mechanism is uniform over all ways of allocating exactly $N_j = 5$ units to each of $J = 4$ treatments:

$$P(\mathbf{W} \mid \mathbf{X}, \mathbf{Y}(1), \dots, \mathbf{Y}(4)) = \begin{cases} \frac{1}{20!/(5!)^4} & \text{if } \sum_{i=1}^{20} \mathbf{1}(W_i = j) = 5 \text{ for all } j = 1, 2, 3, 4, \\ 0 & \text{otherwise.} \end{cases}$$

The number of valid assignments is:

$$|\mathcal{W}^+| = \frac{20!}{5! 5! 5! 5!} = 11,732,745,024.$$

A matched pair design with $P = 10$ pairs has only $2^{10} = 1,024$ valid assignments. The CRD has over ten billion valid assignments compared to just 1,024 — the matched pair design is far more restricted because it enforces balance within every individual pair, not just overall.

(ii) Observed ANOVA decomposition.

The observed total sum of squares decomposes as $SSTotal = SSTreatment + SSResidual$, where:

$$SSTreatment = \sum_{j=1}^4 N_j (\bar{Y}_j^{obs} - \bar{Y}^{obs})^2, \quad SSResidual = \sum_{j=1}^4 \sum_{i: W_i=j} (Y_i^{obs} - \bar{Y}_j^{obs})^2,$$

with $\bar{Y}_j^{obs} = \frac{1}{N_j} \sum_{i: W_i=j} Y_i^{obs}$ the observed mean for group j and $\bar{Y}^{obs} = \sum_j \frac{N_j}{N} \bar{Y}_j^{obs}$ the grand mean. $SSTreatment$ captures how far apart the group means are from the grand mean; $SSResidual$ captures how much individuals vary around their own group mean.

(iii) Group means, grand mean, and sums of squares.

Group means:

$$\begin{aligned} \bar{Y}_1^{obs} &= \frac{3+5+4+6+7}{5} = 5.0, & \bar{Y}_2^{obs} &= \frac{8+10+9+11+7}{5} = 9.0, \\ \bar{Y}_3^{obs} &= \frac{6+8+7+9+5}{5} = 7.0, & \bar{Y}_4^{obs} &= \frac{1+2+3+1+3}{5} = 2.0. \end{aligned}$$

Grand mean:

$$\bar{Y}^{obs} = \frac{5(5.0) + 5(9.0) + 5(7.0) + 5(2.0)}{20} = \frac{115}{20} = 5.75.$$

Treatment sum of squares:

$$\begin{aligned}\text{SSTreatment} &= 5(5.0 - 5.75)^2 + 5(9.0 - 5.75)^2 + 5(7.0 - 5.75)^2 + 5(2.0 - 5.75)^2 \\ &= 5(0.5625) + 5(10.5625) + 5(1.5625) + 5(14.0625) \\ &= 2.8125 + 52.8125 + 7.8125 + 70.3125 = 133.75.\end{aligned}$$

Residual sum of squares (deviations from group mean within each programme):

$$\begin{aligned}\text{SSResidual} &= [(3 - 5)^2 + (5 - 5)^2 + (4 - 5)^2 + (6 - 5)^2 + (7 - 5)^2] \\ &\quad + [(8 - 9)^2 + (10 - 9)^2 + (9 - 9)^2 + (11 - 9)^2 + (7 - 9)^2] \\ &\quad + [(6 - 7)^2 + (8 - 7)^2 + (7 - 7)^2 + (9 - 7)^2 + (5 - 7)^2] \\ &\quad + [(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 + (1 - 2)^2 + (3 - 2)^2] \\ &= (4 + 0 + 1 + 1 + 4) + (1 + 1 + 0 + 4 + 4) \\ &\quad + (1 + 1 + 0 + 4 + 4) + (1 + 0 + 1 + 1 + 1) \\ &= 10 + 10 + 10 + 4 = 34.\end{aligned}$$

$$\text{SSTotal} = 133.75 + 34 = 167.75.$$

(iv) ANOVA table and conclusion.

Source	df	SS	MS	F
Treatments	4 - 1 = 3	133.75	133.75/3 = 44.58	44.58/2.125 = 20.98
Residuals	20 - 4 = 16	34.00	34.00/16 = 2.125	
Total	20 - 1 = 19	167.75		

The observed F -statistic is $F = 20.98$. The critical value from the $F_{3,16}$ distribution at the 5% level is $F_{3,16,0.05} = 3.24$. Since $20.98 \gg 3.24$, we strongly reject the sharp null hypothesis $H_0 : Y_i(1) = Y_i(2) = Y_i(3) = Y_i(4)$ for all i . There is very strong evidence that at least one exercise programme produces a different mean heart rate reduction.

(v) Neymanian inference for $\tau(2, 4)$.

Point estimate:

$$\hat{\tau}(2, 4) = \bar{Y}_2^{\text{obs}} - \bar{Y}_4^{\text{obs}} = 9.0 - 2.0 = 7.0 \text{ bpm.}$$

Within-group sample variances:

$$\begin{aligned}s_2^2 &= \frac{(8 - 9)^2 + (10 - 9)^2 + (9 - 9)^2 + (11 - 9)^2 + (7 - 9)^2}{4} = \frac{1 + 1 + 0 + 4 + 4}{4} = \frac{10}{4} = 2.5, \\ s_4^2 &= \frac{(1 - 2)^2 + (2 - 2)^2 + (3 - 2)^2 + (1 - 2)^2 + (3 - 2)^2}{4} = \frac{1 + 0 + 1 + 1 + 1}{4} = \frac{4}{4} = 1.0.\end{aligned}$$

Conservative estimated variance and standard error:

$$\widehat{\text{Var}}(\hat{\tau}(2, 4)) = \frac{s_2^2}{N_2} + \frac{s_4^2}{N_4} = \frac{2.5}{5} + \frac{1.0}{5} = 0.5 + 0.2 = 0.7, \quad \widehat{\text{SE}} = \sqrt{0.7} \approx 0.837.$$

Approximate 95% confidence interval:

$$\hat{\tau}(2, 4) \pm 1.96 \times \widehat{\text{SE}} = 7.0 \pm 1.96 \times 0.837 \approx 7.0 \pm 1.64 = (5.36, 8.64).$$

The estimator is **conservative** because the exact sampling variance is:

$$\text{Var}(\hat{\tau}(2, 4)) = \frac{S_2^2}{N_2} + \frac{S_4^2}{N_4} - \frac{S_{24}^2}{N},$$

where S_{24}^2 is the variance of individual-level treatment effects $\tau_i(2, 4) = Y_i(2) - Y_i(4)$ across all N units. This term requires observing both $Y_i(2)$ and $Y_i(4)$ for the same unit, which is impossible. Since $S_{24}^2 \geq 0$, dropping it overestimates the true variance, producing confidence intervals that are slightly wider than strictly necessary. This is a safe error: we may be slightly conservative but never systematically undercover.